KODLARIM

design.sv

```
module mux4_8bit (
    input logic [7:0] a,
    input logic [7:0] b,
    input logic [7:0] c,
    input logic [7:0] d,
    input logic [1:0] s,
    output logic [7:0] y
);

    assign y = (s == 2'b00) ? a :
               (s == 2'b01) ? b :
               (s == 2'b10) ? c : d;

endmodule
```

testbench.sv

```
module tb_mux4;

    logic [7:0] a, b, c, d;
    logic [1:0] s;
    logic [7:0] y;

    mux4_8bit u1 (
        .a(a), .b(b), .c(c), .d(d), .s(s), .y(y)
    );

    initial begin
        $dumpfile("wave.vcd");
        $dumpvars(0, tb_mux4);

        a = 8'h12;
        b = 8'h34;
        c = 8'h56;
        d = 8'h78;
        s = 2'b00;

        $monitor("Zaman: %0t | s: %b -> Cikis y: %h", $time, s, y);

        #15 s = 2'b01;
        #15 s = 2'b10;
        #15 s = 2'b11;
        #15 $finish;
    end

endmodule
```

KODLARIN EKRAN ÇIKTISI

```
# KERNEL: Zaman: 0 | s: 00 -> Cikis y: 12
# KERNEL: Zaman: 15 | s: 01 -> Cikis y: 34
# KERNEL: Zaman: 30 | s: 10 -> Cikis y: 56
# KERNEL: Zaman: 45 | s: 11 -> Cikis y: 78
```